Introduction to GPU Programming

Device Global Memory Allocation

Device Memory cudaMalloc Syntax

```
__host__ __device__ cudaError_t cudaMalloc(void **devPtr, size_t size)
```

- Allocates memory to pointer of specified size
- Allocate primitive such as int:

```
int dev_a;
cudaMalloc(&dev_a, sizeof(int));
```

- Allocate arrays:

```
int N = 1024;
int *dev_a;
cudaMalloc(&dev_a, N*sizeof(int));
```

[1]

Device cudaMemcpy Command

- Pageable/Pinned Memory copy (to/from device memory):

```
cudaMemcpy(input_a, dev_a, cudaMemcpyHostToDevice);
cudaMemcpy(dev_a, input_a, cudaMemcpyDeviceToHost);
```

- Mapped memory device memory allocation/copy:

```
cudaHostGetDevicePointer((int **)&dev_a, (int *)input_a, N*sizeof(int));
```

- Unified memory device is allocated on the host so no memory copying just pass the pointer reference to the kernel.

[2]

Bibliography

- [1] 2021. Nvidia. CUDA Toolkit Documentation. Retrieved from https://docs.nvidia.com/cuda/cuda-runtime-api.
- [2] 2017. Nvidia. Unified Memory for CUDA Beginners. Retrieved from https://developer.nvidia.com/blog/unified-memory-cuda-beginners/.

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